Combinatorics

Counting, Ordering and Inclusion

KCPC

March 5, 2026

Overview

Attendance

Our Socials

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Discord

WhatsApp

Why Study Mathematics?

Why Bother?

Purpose Simplify problems and make calculations less computationally intensive.

Motivation

The Locker Problem

Think There are 100 lockers in a high school with 100 students.

$\rightarrow$ The first student opens all lockers

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- → The third student toggles every third locker (closes if open, opens if closed)

The Locker Problem

Think There are 100 lockers in a high school with 100 students.

- → The first student opens all lockers
- → The second student closes all even-numbered lockers $(2, 4, 6, 8 \dots)$
- → The third student toggles every third locker (closes if open, opens if closed)
- → The n-th student toggles every N-th locker

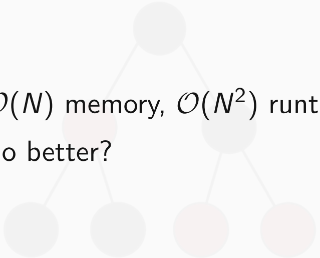
$$(N, 2N, 3N \dots)$$

This continues until all 100 students have had a turn.
Question 1: How many lockers are open at the end of this exercise?
Question 2: What is the general answer for N lockers and N students?

First Thoughts

Naive Solution Store each locker as an 0 entry in an array initially, and then for each number $\leq N$ flip the values from 0 to 1.

Computational Intensity



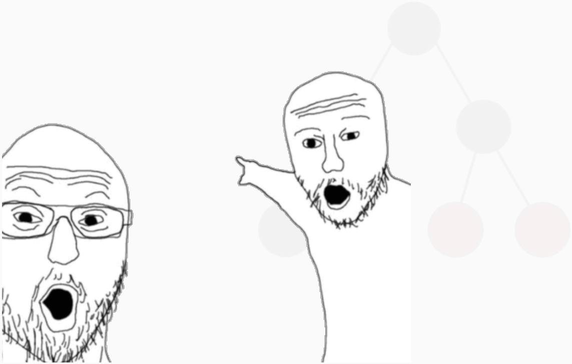
Time Complexity $\mathcal{O}(N)$ memory, $\mathcal{O}(N^2)$ runtime

Inefficient! Can we do better?

Computational Intensity

Yes we can!

$\mathcal{O}(1)$ memory, $\mathcal{O}(1)$ runtime



What is the **key insight**? Going from a computational problem to a logical one

Insight

Solution We are told to consider multiples of N and work from there, so consider hitting every multiple and flipping it.

Change perspectives: what if we consider factors of a general number?

Important idea: Parity

Insight

Solution Each locker is only flipped if the student's number is a (not necessarily proper) factor of it.

The total number of flips is just the total number of factors of the locker's number !

If a number has an even number of factors, it will stay closed at end. If a number has an odd number of factors, it will be open.

Insight

Solution continued Well what numbers have an odd number of factors?

Take a general number. e.g. 18 and consider pairing factors up: $(1,18),(2,9),(3,6)$. Doesn't work, each factor has associated pair.

Try looking at 36: $(1,36),(2,18),(3,12),(4,9),(6,6)$

Only factor that won't be paired up with another number are those paired with themselves.

Only time we have lockers left open is for squares!

For N lockers, just count the number of squares $\leq N$, $\lfloor\sqrt{N}\rfloor$

Insight

Summary Multiple steps required but the solution is elegant.

Key ideas:

- → Parity
- → Pairing
- → Testcases
- → Factorisation

All these are fundamental tools!

How do you get better at spotting what's useful where?

DO PROBLEMS

Basic Counting Techniques

Introduction to Counting

Motivation Any toddler on the street (and hopefully you) can count the number of apples in a line.

Introduction to Counting

Motivation

But do you know off the top of your head:

- → How many full houses are possible in a standard deck of cards?
- → How many possible phone codes using a 6-digit PIN?
- → How many credit card numbers are possible using Luhn checksum?
- → How many shortest paths from one side of a chess board to another?
- → How many necklaces you can make from 5 red beads and 20 black ones?

Counting Revision

Multiplication Principle

I have 5 possible breakfast meals, 10 possible lunch meals and 20 possible dinner meals.

How many possible combinations of meals can I have?

Counting Revision

Multiplication Principle

You already have the informal idea that if we have a process with k independent steps, and $n_1, n_2, \ldots, n_k$ choices at each step, we have $n_1 n_2 \ldots n_k$ outcomes!

This is Holy Grail of counting, problem is figuring out those n_k, especially when they are related.

Counting Revision

Pigeonhole Principle

I have $k + 1$ pigeons out delivering parcels for me. I only have k pigeonholes for them to roost in when they come back. Can every pigeon have their own nest?

Counting Revision

Pigeonhole Principle NO!

Counting Revision

Pigeonhole Principle If we put $mn + 1$ pigeons into n pigeonholes, then some pigeonhole has at least $m + 1$ pigeons.

Counting Revision

Pigeonhole Principle

How many people are there in the room right now? How many do I need to pick out so I can guarantee they have the same birth month? The same birthday? The same birth year?

Factorials

Ordering

How many ways can I make Anas, Beed, Cutay, Datthew and Ercell sit in a line?

Factorials

Ordering

How many ways can I make Anas, Beed, Cutay, Datthew and Ercell sit in a line?

5 possible choices for A, 4 for B, 3 for C, 2 for D and 1 for E.

So $5 * 4 * 3 * 2 * 1 = 5! = 120$ in total. These are **factorials** and show up everywhere.

Computers and Counting

Compilation

Try computing the number of arrangements of a deck of card, we have

$$52! = 8.0658175 * 10^{67} = 8065817517094387857166063685640376697528$$

Python can compute it in milliseconds since it works with arbitrary precision.

Java, C and C++ actually overflow at around 10^{21}.

Computers struggle with counting!

Picking Things

Non-Ordered

We have our friends Anas, Beed, Cutay, Datthew, Ercell and Femeth again.

How many ways can we pick exactly 3 of our friends to hang out with?

Picking Things

Non-Ordered

We have our friends Anas, Beed, Cutay, Datthew and Ercell again.

How many ways can we pick exactly 3 of our friends to hang out with?

This is another chance to change our perspective! Think about ordering these 6 in a line again, but let's say you only pick the first 3 to hang out with, and the other 3 make the next workshop.

Picking Things

Non-Ordered

We have $5! = 120$ ways to order our first 3 friends, but note that Anas, Beed, Cutay is the same group of friends as Cutay, Beed, Anas. So our estimate includes possibilities we have already considered: we have **double-counted**!

So we have to divide our total by a factor of 3!.

For the other friends, yet again Datthew, Ercel is the same group of workers as Ercel, Datthew, so we need to divide our total by a factor of 2!.

In total, we have exactly

$$\frac{5!}{3! * 2!} = 10$$

unique ways to pick 3 from 5.

The Choose Function

In total, we have exactly

$$\frac{5!}{3! * 2!} = 10$$

unique ways to pick 3 from 5. This is also called the **binomial coefficient** $\binom{5}{3}$ or **choose function**, ${}^5C_3$.

In general, if we want to pick r things from a group of n, $r \leq n$, we have

$$\binom{n}{r} = \frac{n!}{r!(n-r)!} = {}^n C_r$$

The Choose Function

Binomial Identities Eagle-eyed among you may have noticed that $\binom{n}{r} = \binom{n}{n-r}$ algebraically.

We can also think of this as the number of ways to pick r elements from a set of n elements is the same number of ways to pick $n - r$ to not be included.

Pascal's Triangle

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1
1 7 21 35 35 21 7 1

symmetric

Note the symmetry: $\binom{n}{r} = \binom{n}{n-r}$

Worked Example

There are exactly ten ways of selecting three from five, 12345:

123, 124, 125, 134, 135, 145, 234, 235, 245, and 345

In combinatorics, we use the notation, $\binom{5}{3} = 10$.

In general, $\binom{n}{r} = \frac{n!}{r!(n-r)!}$, where $r \leq n$, $n! = n \times (n-1) \times \ldots \times 3 \times 2 \times 1$, and $0! = 1$.

It is not until $n = 23$, that a value exceeds one-million: $\binom{23}{10} = 1144066$.

How many, not necessarily distinct, values of $\binom{n}{r}$ for $1 \leq n \leq 100$, are greater than one-million?

Worked Example

Naive approach is to test each value separately, this gives 5050 in total, and is slow to compute.

Take a step back and think: have symmetry between r and $n - r$, so only need to compute half the values.

Looking closer: each value is the sum of previous terms.

So can set up some 2D array in a DP-style, and if any term hits 1000000, we know the sub-triangle below it is all above 100000 too!

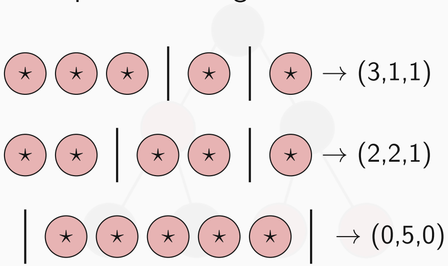
Worked Example

Pseudocode

```
COUNT = 0
C = [[0] * 101 for _ in range(101)]
for n in range(101):
    C[n][0] = 1
    for r in range(1, n // 2 + 1):
        C[n][r] = C[n-1][r-1] + C[n-1][r]
        if C[n][r] > 1000000:
            C[n][r] = float('inf')
            COUNT += 1
        C[n][n-r] = C[n][r]
print(COUNT)
```

A New Star

Very soon you'll run into problems like how many ways can 5 identical sweets be split into 3 bags?



Choosing which 2 of 7 positions are bars: $\binom{7}{2} = 21$

Stars and Bars

General Formula

Distributing n identical items into k distinct bins:

$$\binom{n+k-1}{k-1}$$

Why? We have n stars and $k-1$ bars — $n+k-1$ symbols total. We choose which $k-1$ positions are bars.

Note: this allows empty bins. If each bin must have ≥ 1 item, give one to each bin first, then distribute $n-k$ items freely:

$$\binom{n-1}{k-1}$$

Bijection and Proofs

Dreamland

Wouldn't it be nice if instead of counting something we could count the elements of another set that just so happens to be the exact same size?

That is the motivation behind **bijections:** one-to-one mappings.

Definition

Formally, a bijection between two sets A, B is a function $f : A \to B$ such that:

- $\to$ **Injectivity** each element in A maps to a different element in B, $f(a_i) = f(a_j) \Rightarrow a_i = a_j$
- $\to$ **Surjectivity** each element in B is mapped to by an element in A

Key idea: if A, B are in bijection, $|A| = |B|$.

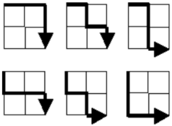
If we want to count A, but B is a lot easier, we're happy ☺

Worked Example

Lattice Paths

Problem 15

Starting in the top left corner of a 2×2 grid, and only being able to move to the right and down, there are exactly 6 routes to the bottom right corner.



How many such routes are there through a 20×20 grid?

Worked Example

Can accomplish this purely mathematically!

Instead of counting paths (you will not get through them in a reasonable amount of time), call each move right an R and each move down a D. Then **each path has an associated string!**

Every string with 20 R's and D's covers all valid paths.

Worked Example

Try doing this with order, say you have 20 R's in a row and try to insert D's: risk over-counting.

How do we do this properly?

Ignore ordering, consider every possible arrangement of the 40 R's and D's, have 40! in total. Each R and each D are identical, so need to divide by a factor of 20!, get

$$\frac{40!}{20! * 20!} = 137846528820$$

Double-Counting

A related technique: count the same set two different ways to derive an identity.

Example: Prove that $\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n$

- $\rightarrow$ **Left side:** sum over all possible subset sizes — counts every subset of $\{1, \ldots, n\}$ exactly once
- $\rightarrow$ **Right side:** each element is either in or out, giving 2^n subsets total
- $\rightarrow$ Same set, counted two ways $\implies$ they must be equal $\square$

Double-Counting Example

Claim: n points on a circle, every pair connected, no three chords concurrent inside. The number of interior intersections is $\binom{n}{4}$

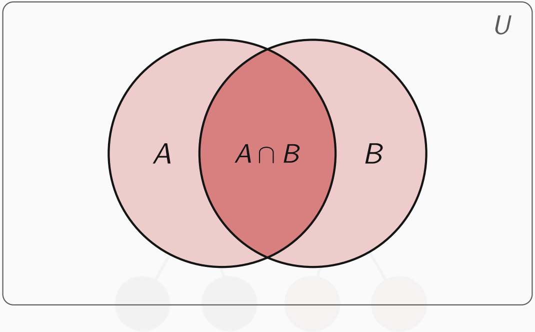
Inclusion-Exclusion

Why Bother

Motivation: If I need to count the number of events in A and B, what do I need to do to ensure that I haven't double counted anything?

What about for 3 sets? Or more?

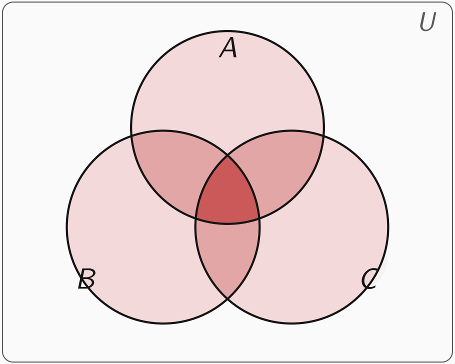
Venn Diagrams



When we add $|A| + |B|$, we count $A \cap B$ **twice**. So:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

3-Circle Diagram



Three Sets

Adding $|A| + |B| + |C|$ overcounts pairwise intersections and undercounts the triple intersection. Fixing this gives:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

General pattern: add individuals, subtract pairs, add triples, subtract quadruples ...

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{i} |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots$$

Derangements

Application: How many permutations of $\{1, \ldots, n\}$ have **no fixed points**?

Let A_i = permutations where element i is in position i. We want $\left|\overline{A_1} \cap \cdots \cap \overline{A_n}\right|$.

By PIE:

$$D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}$$

- → $D_1 = 0, \quad D_2 = 1, \quad D_3 = 2, \quad D_4 = 9$
- → As $n \to \infty$, $D_n/n! \to e^{-1} \approx 0.368$

Intuition: roughly $1/e$ of all permutations are derangements regardless of n.

Extension

Further Directions

Catalan Numbers

- → $C_n = \dfrac{1}{n+1}\dbinom{2n}{n}$ counts many things like bracket sequences and non-crossing grid paths
- → $C_0 = 1, C_1 = 1, C_2 = 2, C_3 = 5, C_4 = 14$

Burnside's Lemma

- → Counts distinct objects under symmetry — e.g. necklaces, colourings
- → Requires group theory (see Groups notes in reading list)
- → Answer to the necklace problem from the start of the session!

Full treatment in the supplementary reading document.

Problem Sheet

Check it out online!

→ Problems available on the KCPC Discord and website

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- → We will review selected solutions afterwards

Final Remarks

This may be annoying at first but do keep trying!

- $\rightarrow$ Multiplication principle — independent stages multiply
- $\rightarrow$ Factorials and binomial coefficients — ordered vs unordered
- $\rightarrow$ Stars and bars — distributing identical items
- $\rightarrow$ Bijections — count something easier instead
- $\rightarrow$ Inclusion-exclusion — correct for overcounting

These will come up constantly. The only way to get faster at spotting them is to **do problems**.

Further Reading

→ A document containing relevant supplementary + research reading will be available on the KCPC website (for this workshop) and the KCPC Discord Resource section for you to check out :D